

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI**

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: ONE

CLASS: SS3

SUBJECT: Agricultural Science

TOPIC: Fisheries & Fish Farming

Introduction

- Fisheries is the science of catching, rearing and processing fish and other aquatic animals. Fish farming (aquaculture) is the cultivation of fish in ponds, tanks, cages or raceways under controlled conditions.

Importance of fisheries

- Fish is a cheap source of animal protein; fishing provides employment for coastal and riverine communities; it earns foreign exchange (shrimps, frozen fish exports) and replaces imported fish; fish scales, oil and offal feed livestock; it makes use of water bodies.

Types and systems of fish farming

- **Extensive** — natural water bodies, minimal inputs; **semi-intensive** — ponds with feeding and stocking; **intensive** — high stocking with complete feeds and aeration (earthen ponds, concrete tanks, cages in rivers/lakes, and fibre tanks). Common species in Nigeria: tilapia, catfish (*Clarias*, *Heterobranchus*), carp, mudfish.

Managing a fish farm

- Site: good water supply, clay soil, flat land near a market, away from flooding. Pond preparation: clear and lime the pond, fertilise to grow plankton (natural feed), stock fingerlings at the right density, feed with floating pellets (5% of body weight daily), control predators and poachers, check water quality (pH 6.5–8.5, oxygen), harvest after 4–6 months with a net, then process fish (cleaning, smoking, freezing) before marketing.

CLASS EVALUATION

1. Define fisheries and aquaculture.
2. State three importance of fish farming.
3. List four species of fish reared in Nigeria.
4. Describe how you would prepare a fish pond before stocking.

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: TWO

CLASS: SS3

SUBJECT: Agricultural Science

TOPIC: Apiculture & Snail Rearing

Introduction

- Apiculture is the keeping of bees (for honey, beeswax and pollination); snail rearing is heliculture — breeding snails for meat. Both are low-cost, small-space enterprises.

Bee keeping

- The hive: modern Langstroth hives give cleaner honey (movable frames); the colony has a queen (egg layer), workers (foraging, building, guarding) and drones (mating). Equipment: hive, smoker (calms bees), bee brush, veil and gloves, honey extractor. Siting: near flowers, away from people; facing away from footpaths; on stands. Management: inspect regularly, control pests (wax moth) and diseases, avoid disturbing in bad weather, provide water, harvest honey when capped (7–8% moisture), leave enough for the colony.

Products and importance

- Honey (food, medicine, sweetener), beeswax (candles, cosmetics, polish), propolis, royal jelly. Bees pollinate crops, raising yields; apiculture gives income and employment with little land.

Snail rearing

- Snails need a cool, moist, shaded pen with humus soil, calcium (for shells) and a fence (they climb); feed on leaves (cabbage, pawpaw, waterleaf), fruits and snail feed; control predators (rats, ants); harvest at maturity (about 8–12 months); process and market. Species: *Archachatina marginata* (giant land snail).

CLASS EVALUATION

1. Define apiculture.
2. State the three castes of a bee colony and their functions.
3. List two products of the hive.
4. What conditions and feed do snails need?

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: THREE

CLASS: SS3

SUBJECT: Agricultural Science

TOPIC: Pasture & Forage Crops

Introduction

- Pasture is land covered with grass and other plants grazed by livestock; forage crops are plants grown to be cut and fed to animals (hay, silage, fresh fodder). They are the cheapest source of feed for ruminants.

Types of pasture

- **Natural pasture** — unimproved grassland (woody shrubs and grasses); low in nutrients, seasonal, low carrying capacity. **Improved pasture** — planted with good grasses and legumes, fertilised, fenced and rotationally grazed; higher yields. **Rotational grazing** divides the pasture into paddocks grazed in turn, allowing rest and regrowth; **zero grazing** cuts and carries fodder to the animals.

Common pasture grasses and legumes

- Grasses: Panicum maximum (guinea grass), Cynodon dactylon (Bermuda grass), Andropogon gayanus (gamba grass), Brachiaria species, Pennisetum (elephant grass). Legumes: Centrosema, Stylosanthes, Pueraria (kudzu) and Leucaena. Legumes fix nitrogen, improving soil fertility and feed protein.

Conservation of forage

- **Hay** — grass dried in the sun until brittle, then stored dry (moisture below 20%). **Silage** — chopped green forage packed and fermented anaerobically in a pit or bag; the acid preserves it, best from maize or grass mixed with a little molasses. Hay for dry season roughage; silage keeps more nutrients.

CLASS EVALUATION

1. Define pasture and forage.
2. Differentiate between natural and improved pasture.
3. Name two legumes used in pastures and their value.
4. How is silage made?

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: FOUR

CLASS: SS3

SUBJECT: Agricultural Science

TOPIC: Farm Produce Processing & Preservation

Introduction

- Farm produce is perishable; processing and preservation prevent spoilage, add value, extend shelf life and stabilise prices. Methods are physical, chemical and biological.

Why process and preserve

- Reduces post-harvest losses (25–40% in Nigeria); makes produce available all year; adds value and income (garri from cassava, flour from wheat); improves quality, safety and transportability; creates jobs.

Methods of preservation

- **Drying/sun-drying** — fish, pepper, maize, cassava chips; **smoking** — fish (heat and smoke kill microbes); **salting** — fish and meat (salt draws out water); **refrigeration/freezing** — meat, fish, vegetables; **canning/bottling** — heat-sealed against microbes (tomato paste, fruits); **fermentation** — cassava into garri, milk into yoghurt, oil palm sap into wine; **chemical preservation** — salt, sugar, vinegar, sodium benzoate; **irradiation** and **pasteurisation** (milk) for longer life.

Processing common Nigerian crops

- Cassava → garri, fufu, starch, cassava flour; oil palm → palm oil, kernel oil, soap; cocoa → cocoa butter, chocolate; groundnut → oil, cake, butter; maize → flour, ogi, popcorn; rubber → latex, crepe. Processing requires clean water, hygienic handling and good storage (dry, ventilated, pest-free stores).

CLASS EVALUATION

1. State two reasons for preserving farm produce.
2. Differentiate between smoking and canning.
3. How is garri made?
4. Mention three chemical preservatives.

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: FIVE

CLASS: SS3

SUBJECT: Agricultural Science

**TOPIC: Agricultural Marketing & The Farmer's Business
(Revision)**

Introduction

- Agricultural marketing is the process of moving farm products from the farm to the final consumer — assembling, grading, processing, packaging, transporting and selling. This week revises the business side of farming.

Marketing channels and functions

- Channels: farmer → local market → wholesaler → retailer → consumer (long chains, many middlemen). Functions: assembling (bulking produce), grading and standardising, processing, storage, packaging, transportation, financing, risk bearing and price discovery. The middleman takes a big share of the consumer's price — farmers get less than 50% in many cases.

Problems of agricultural marketing

- Poor storage and processing facilities; poor rural roads; price instability and seasonality; middlemen exploitation; lack of market information; inadequate credit; adulteration; poor packaging and grading.

Farm business management

- Planning (set goals, budget, choose enterprises); records (inventory, production, sales, cash book); costing (fixed and variable costs; break-even point); pricing; risk management (insurance, diversification, safe storage); and marketing strategy (cooperatives, guaranteed minimum price, commodity boards historically). Cooperative societies pool produce for better prices and bulk buying.

CLASS EVALUATION

1. Define agricultural marketing.
2. List five marketing functions.
3. State four problems of agricultural marketing.
4. How can a cooperative help the farmer?